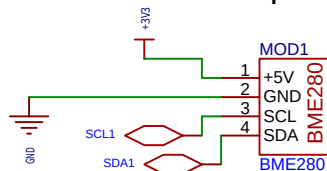


Modulo Sensor de Temperatura y Humedad

Diagrama de conexión para el módulo sensor BME280:

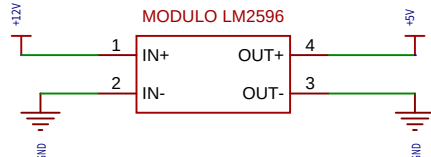
- Pin 1 (+3V3) se conecta a +3V3.
- Pin 2 (+5V) se conecta a +5V.
- Pin 3 (GND) se conecta a GND.
- Pin 4 (SCL) se conecta a SCL1.
- Pin 5 (SDA) se conecta a SDA1.

El módulo está etiquetado como MOD1 y BME280.

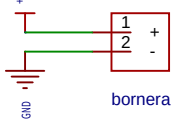


Step Down 12V a 5V

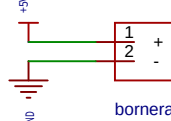
The diagram shows an LM2596 voltage regulator in a TO-18 package. The input is a +12V supply connected to pin 1 (IN+). The input is also connected to pin 2 (IN-), which is then connected to a ground symbol. The output is taken from pin 4 (OUT+), which is connected to a +5V supply. Pin 3 (OUT-) is connected to a ground symbol. The text 'MODULO LM2596' is written above the component.



Modulo de Bornera1 (Fuente 12v)

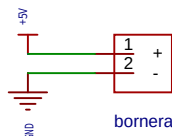


Modulo de Bornera2 (Salida Ventilador)



Modulo de Bornera3 (Salida Peltier)

The diagram shows a two-pin connector labeled 'bornera'. Pin 1 is connected to a +5V supply, and pin 2 is connected to GND. The connector has a red '+' sign on pin 1 and a black '-' sign on pin 2.



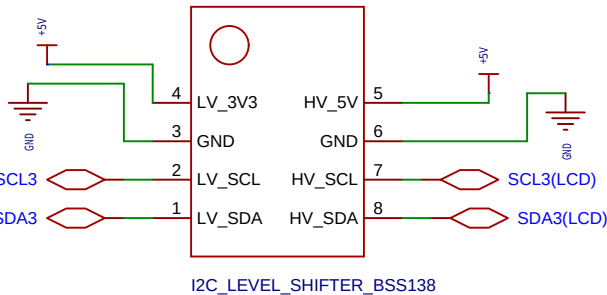
Modulo Level Shifter 12C

The diagram illustrates the wiring for the Modulo Level Shifter 12C module. The module is shown as a central component with two sets of pins: LV (Low Voltage) and HV (High Voltage). The LV pins are connected to a 3.3V supply, and the HV pins are connected to a 5V supply. The LV_SCL and LV_SDA pins are connected to the SCL3 and SDA3 pins of the LCD module, respectively. The HV_SCL and HV_SDA pins are connected to the SCL3(LCD) and SDA3(LCD) pins of the LCD module, respectively. The GND pins are connected to ground.

Pin Connections:

- LV_3V3 (Pin 4) → 3.3V
- LV_GND (Pin 3) → GND
- LV_SCL (Pin 2) → SCL3
- LV_SDA (Pin 1) → SDA3
- HV_5V (Pin 5) → 5V
- HV_GND (Pin 6) → GND
- HV_SCL (Pin 7) → SCL3(LCD)
- HV_SDA (Pin 8) → SDA3(LCD)

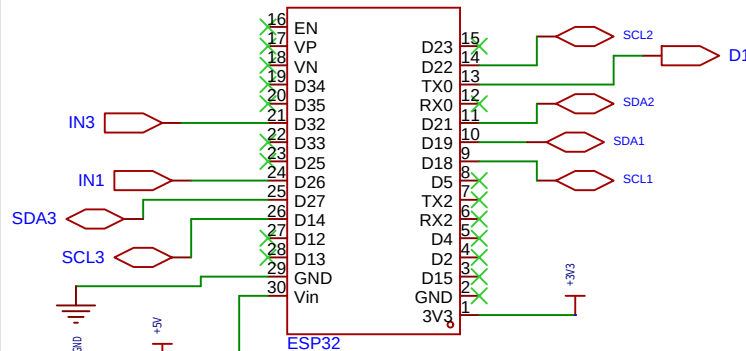
Module Label: I2C_LEVEL_SHIFTER_BSS138



Modulo Principal

The diagram illustrates the wiring of an ESP32 module. The module's pins are connected as follows:

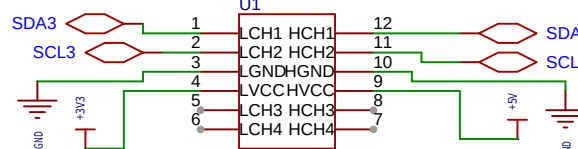
- Power and Ground:**
 - EN (16) is connected to +5V.
 - VP (17), VN (18), D25 (24), D26 (25), D27 (26), D14 (27), D12 (28), D13 (29), GND (30), and Vin are connected to GND.
 - 3V3 (1) is connected to +3V3.
- Signal Connections:**
 - D34 (20) is connected to IN3.
 - D35 (21) is connected to IN1.
 - D32 (22) is connected to SDA3.
 - D33 (23) is connected to SCL3.
 - D23 (15) is connected to SCL2.
 - D22 (14) is connected to D1.
 - RX0 (12) is connected to SDA2.
 - D21 (11) is connected to SDA1.
 - D19 (10) is connected to SCL1.
 - D18 (9), D5 (8), TX2 (7), RX2 (6), D4 (5), D2 (4), D15 (3), and GND (2) are connected to GND.



Modulo de Level Shifter

The diagram shows a Level Shifter module (U1) with the following connections:

- Pin 1 (LCH1):** Connected to +3V3.
- Pin 2 (LCH2):** Connected to GND.
- Pin 3 (LCH3):** Connected to +3V3.
- Pin 4 (LCH4):** Connected to GND.
- Pin 12 (HCH1):** Connected to +5V.
- Pin 11 (HCH2):** Connected to GND.

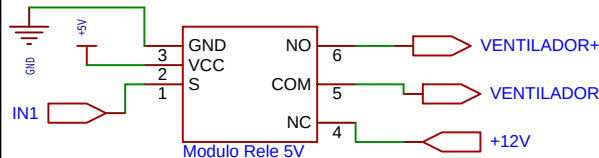


Modulo Relé Ventilador

The diagram shows a 5V relay module with the following connections:

- VCC (Pin 3):** Connected to a 5V power source.
- GND (Pin 1):** Connected to ground.
- S (Pin 2):** Connected to a switch labeled IN1.
- NO (Pin 6):** Connected to the positive terminal of a fan, labeled VENTILADOR+.
- COM (Pin 5):** Connected to the negative terminal of a fan, labeled VENTILADOR-.
- NC (Pin 4):** Connected to a 12V power source, labeled +12V.

The module is labeled "Modulo Relé 5V".

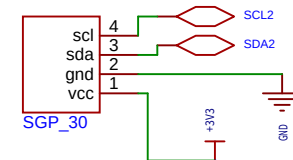


Modulo Sensor de Calidad de Aire

The diagram shows the SGP-30 sensor module connected to an I2C bus. The sensor's pins are connected as follows:

- Pin 1 (vcc) to +3V3
- Pin 2 (gnd) to GND
- Pin 3 (sda) to SDA2
- Pin 4 (scl) to SCL2

The SDA2 and SCL2 lines are terminated with resistors.



Modulo Pantalla LCD

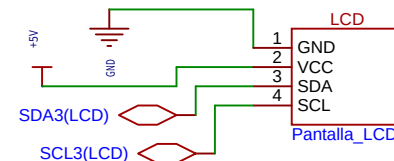
Diagrama de conexión para el módulo Pantalla LCD:

- Pin 1: GND
- Pin 2: VCC
- Pin 3: SDA
- Pin 4: SCL

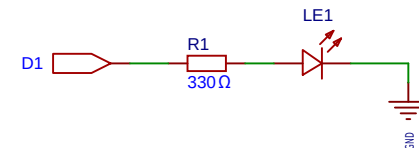
Conexiones:

- Pin 1 (GND) conectado a tierra.
- Pin 2 (VCC) conectado a +5V.
- Pin 3 (SDA) conectado al pin SDA3(LCD) del módulo.
- Pin 4 (SCL) conectado al pin SCL3(LCD) del módulo.

El módulo se etiqueta como **Pantalla_LCD**.

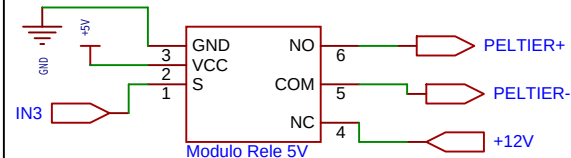


Modulo de Led Indicator



Modulo Rele Celda Peltier

The diagram illustrates the wiring for a Peltier cell relay module. A 5V power source is connected to the GND, VCC, and S pins of the module. The NO pin is connected to the Peltier+ terminal, the COM pin to the Peltier- terminal, and the NC pin to the +12V terminal. A blue label 'Modulo Rele 5V' is positioned below the module.



Schematic	ESQUEMATICO_ECOFRESH			Create at	2026-05-26
				Update at	2026-06-10
Board	Board1			Page	ECOFRESH1
Drawn	Equipo_05	ECOFRESH			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	